Quantum Field Theory – Homework 1

Name: ______________________ **Date:** _____________

The bullet points under each problem are structured hints or guiding points to help you know what to include in your solution.

Problem 1. Classical Field Theories and Principle of Locality (20 points)

Summarize the key points of **Lecture 1** on *Classical Field Theories and the Principle of Locality* and **Lecture 2** on *Symmetries and Conservation Laws.*

- Discuss the motivation for introducing field theory from classical mechanics.
- Explain how the **principle of locality** restricts physical interactions.
- Describe how **continuous symmetries** lead to **conservation laws** through **Noethers theorem**.

Problem 2. Derivation of Equations of Motion in Classical Mechanics (20 points)

Summarize the general steps to derive the **equations of motion** in classical mechanics.

- Start from the action principle $S[q] = \int L(q, \dot{q}, t)\, dt$.
- Show how the variation of the action leads to the **Euler-Lagrange equation**.
- State the assumptions (fixed endpoints, small variations, etc.).

Problem 3. Equation of Motion in Classical Field Theory (20 points)

Repeat the process to obtain the **equation of motion** in classical field theory for the potential

$$V(\phi) = \frac{1}{2}m^2\phi^2.$$

Provide detailed derivations.

- Start from the field action $S[\phi] = \int d^4x\, \mathcal{L}$.
- Compute the variation of the Lagrangian density and derive the Euler-Lagrange field equation.
- Show that this gives the **KleinGordon equation**:

$$(\Box + m^2)\phi = 0.$$

Problem 4. Translation Symmetry of the Action (20 points)

Prove in detail that the **action is invariant** under spacetime translations

$$x^\mu \to x'^\mu = x^\mu + a^\mu.$$

- Compute the change in the field $\delta\phi(x)$.
- Show explicitly that $\delta S = 0$.
- Derive the **energy-momentum tensor** from Noethers theorem.

Problem 5. Lorentz Symmetry of the Action (20 points)

Prove that the **action remains invariant** under Lorentz transformations

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}.$$

- Compute the transformation of the field and its derivatives.
- Show that the scalar field Lagrangian

  $$\mathcal{L} = \frac{1}{2}(\partial_{\mu}\phi)(\partial^{\mu}\phi) - V(\phi)$$

  is Lorentz invariant.
- Identify the corresponding **conserved angular momentum tensor**.